CitizenSense - NLP-Based Public Feedback Classification and Insight System

AIRI Team PITB - AI Internship Task 4 | DRAFT: rewrite in your own words and add your transformer results, screenshots and name.

1. Overview

CitizenSense analyses short public feedback (English, Roman Urdu, mixed) and returns category, sentiment, priority, keywords, a summary and a suggested department. It is a learning prototype; predictions are suggestions, not decisions.

2. Data

The dataset is **synthetic**, generated from hand-written phrase banks with injected noise. 1,512 raw rows became 1,251 after removing empty rows, 253 exact duplicates and 1 near-duplicate. 8 categories, split 70/15/15 (875/188/188) stratified by category. Cleaning: lowercasing, URL/phone/CNIC/email/emoji removal, punctuation and whitespace handling, typo/Roman Urdu normalisation, label encoding. A separate 49-message hand-written challenge set was kept out of training.

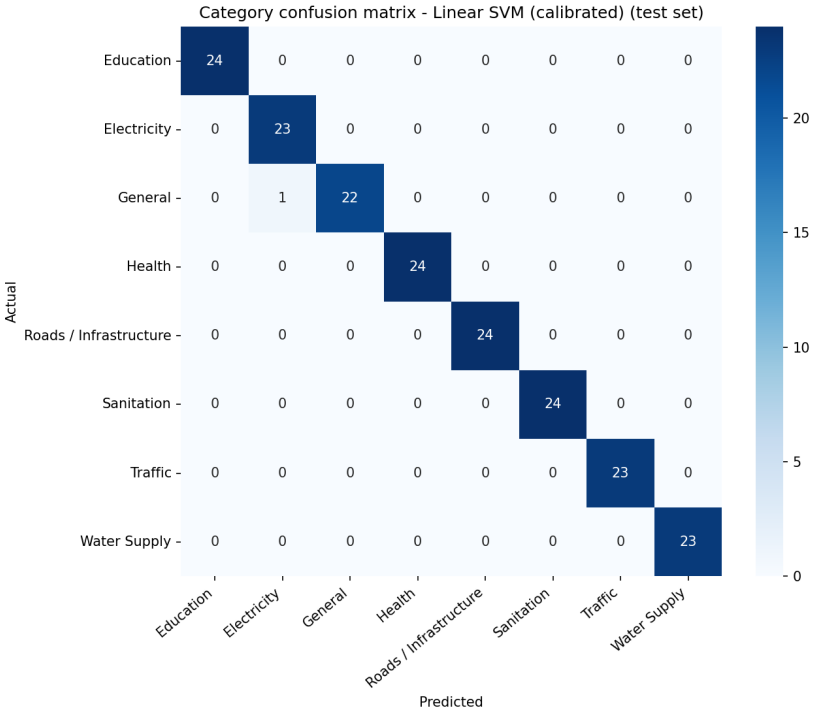
3. Methodology

TF-IDF word (1-2) and character (2-5) n-gram features; Naive Bayes, Logistic Regression, calibrated Linear SVM and Random Forest compared on validation macro-F1; best model evaluated once on test. Keywords use TF-IDF with a domain lexicon boost; summaries are template-based. Hugging Face components (zero-shot, multilingual sentiment, DistilBART, MiniLM embeddings) are implemented in app/hf_models.py. [Add your Colab transformer results here.]

4. Results

Task	Model	Val F1	Test Acc	Test F1
category	Naive Bayes	0.967	0.968	0.968
category	Logistic Regression	0.973	0.984	0.984
category	Linear SVM (calibrated)	0.984	0.995	0.995
category	Random Forest	0.984	0.979	0.978
sentiment	Naive Bayes	1.0	0.989	0.986
sentiment	Logistic Regression	1.0	1.0	1.0
sentiment	Linear SVM (calibrated)	1.0	1.0	1.0
sentiment	Random Forest	0.986	0.989	0.987
priority	Naive Bayes	0.649	0.676	0.67
priority	Logistic Regression	0.73	0.787	0.784
priority	Linear SVM (calibrated)	0.7	0.745	0.741
priority	Random Forest	0.698	0.766	0.76

Category on the template test split: accuracy 0.995. On the 49 hand-written challenge messages: accuracy 0.592, macro-F1 0.551 (best: Education; weakest: Traffic and Water Supply). Accuracy alone can mislead because sentiment is about 73% Negative; macro-F1 weighs every class equally. The gap between the two category scores shows the template split is optimistic and that more diverse data is the main improvement.


Category confusion matrix - Linear SVM (calibrated) (test set)
Education
Electricity
General
Health
Roads / Infrastructure
Sanitation
Traffic
Water Supply
Actual
Predicted
24 0 0 0 0 0 0 0
0 23 0 0 0 0 0 0
0 1 22 0 0 0 0 0
0 0 0 24 0 0 0 0
0 0 0 0 24 0 0 0
0 0 0 0 0 24 0 0
0 0 0 0 0 0 23 0
0 0 0 0 0 0 0 23
20
15
10
5
0

5. Error Analysis (sample)

Text	Actual	Predicted	Type
The park is dirty and the street lights are off	General	Electricity	Wrong category
Janab, hamare muhalle mein koi dustbin nahi hai.	Sanitation	Sanitation	Correct but low confidence
hamare muhalle mein koi dustbin nahi hai. Please look into it.	Sanitation	Sanitation	Correct but low confidence
School has no water supply and the toilets are dirty	Education	Education	Correct but low confidence
school ki road par traffic jam rehta hai	Traffic	Traffic	Correct but low confidence
toot-phoot road par barish ka pani jama ho jata hai for a week. This is an emergency, act immediately!	Roads / Infrastructure	Roads / Infrastructure	Correct but low confidence
I want to report that overloaded rickshaws and buses are a danger to the public in our locality for a week. Pl	Traffic	Traffic	Correct but low confidence
There is a dead animal on the road and nobody removes it near the main market.	Sanitation	Sanitation	Correct but low confidence
Illegal construction is going on in the residential area.	General	General	Correct but low confidence
Sir, Shops have encroached on the footpath in our locality.	General	General	Correct but low confidence
Complaint: Illegal construction is going on in the residential area.	General	General	Correct but low confidence
zyada sawariyan bithanay wale rickshaw aur bus khatarnak hain kai hafton se.	Traffic	Traffic	Correct but low confidence
Complaint: The government office staff behaved badly with us near the main market for months. Kindly take acti	General	General	Correct but low confidence
Meherbani karke note karein, bijli ka bill bohat zyada aya hai do din se. Meherbani karke dekhein.	Electricity	Electricity	Correct but low confidence
Overloaded rickshaws and buses are a danger to the public near the main market. Barae meherbani karwai karein.	Traffic	Traffic	Correct but low confidence
Sir, park ke saath kachra phenkne se beemariyan phail rahi hain teen din se.	Sanitation	Sanitation	Correct but low confidence
park mein benches aur roshni ki zaroorat hai teen din se.	General	General	Correct but low confidence
The speed breaker is too high and damages vehicles in Block C for a week. It is a serious problem for everyone	Roads / Infrastructure	Roads / Infrastructure	Correct but low confidence

Improvements: (1) add varied hand-written samples, especially short, Roman Urdu and multi-issue text; (2) fine-tune a multilingual transformer; (3) multi-label classification; (4) human review for low-confidence predictions.

6. Limitations

Synthetic data; rule-derived priority labels; unstable Neutral/Positive metrics (small classes); inconsistent Roman Urdu spelling; not suitable for real decision-making.

7. What I Learned

[Write 5-7 bullets in your own words.]